Gradient of Knowing (GoK-8): An Integrative Learning-Progression Model and Conceptual Assessment Framework

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Abstract

This paper proposes the Gradient of Knowing (GoK-8), an integrative conceptual framework for interpreting evidence of knowing across learning, practice and contribution. GoK-8 distinguishes eight descriptors: Unawareness, Knowledge, Understanding, Controlled Tryout / Demonstration, Accountable Practice, Mastery, Creative / Adaptive Expertise, and Founding / Field-Building. The framework synthesizes established traditions in educational objectives, competence assessment, skill acquisition, structural learning outcomes, adaptive expertise, learning progressions and rubric-based assessment.

The central contribution is not a claim of empirical validation. Rather, GoK-8 is presented as a diagnostic map and rubric candidate for distinguishing forms of evidence that are often collapsed under the single term "knowing." The model is especially concerned with the difference between controlled demonstration and accountable real-world practice, and with the difference between routine mastery and adaptive or creative performance. Stage-model critiques require caution: GoK-8 should not be interpreted as a deterministic universal ladder. Its levels are intended as heuristic, domain-specific and empirically revisable descriptors. The eighth descriptor, Founding / Field-Building, is retained as a provisional trans-individual extension because it depends on adoption by others rather than individual competence alone.

The paper concludes with a validation agenda. Future work should include expert review, artifact sampling in a concrete pilot domain, rater training, inter-rater reliability assessment, comparison to competing models and evaluation of instructional usefulness. Scientific-writing development is proposed as a suitable first pilot domain because it produces observable artifacts across multiple levels: notes, literature matrices, outlines, drafts, revisions and submission packages.

Keywords: learning progression; competence assessment; rubric; mastery; adaptive expertise; professional development; scientific writing; conceptual framework.

1. Introduction

Educational and professional-development systems regularly face a practical question: what does it mean for a person to know something well enough to act with it? The answer is rarely simple. A person may recognize terminology without understanding relationships. Another may explain a concept clearly but fail to perform a task under real constraints. A learner may succeed in a simulation but not in accountable practice. A practitioner may perform reliably in routine cases but struggle when the situation changes. An expert may produce novel work, but novelty alone does not guarantee adaptive quality.

Despite these distinctions, educational language often treats "knowing" as a single state. This creates assessment errors. A quiz score can be mistaken for competence. A demonstration can be mistaken for real practice. A single successful performance can be mistaken for mastery. A novel output can be mistaken for creativity even when it lacks quality or fit. These confusions matter in schools, universities, professional training, workplace development and scientific writing.

The Gradient of Knowing (GoK-8) is proposed as a practical framework for separating these states. It uses eight descriptors:

1. Unawareness
2. Knowledge
3. Understanding
4. Controlled Tryout / Demonstration
5. Accountable Practice
6. Mastery
7. Creative / Adaptive Expertise
8. Founding / Field-Building

GoK-8 is not intended to replace existing models. It is intended to integrate several partial perspectives into one diagnostic language. Bloom's taxonomy and its revision help formulate cognitive objectives. Miller's pyramid helps distinguish knowing, showing and doing. Skill-acquisition models describe movement toward fluent performance. SOLO taxonomy highlights the structural quality of understanding. Adaptive-expertise literature distinguishes routine fluency from flexible innovation. Learning-progression and rubric literature define the standard for future validation.

The research question guiding this paper is: Can an eight-level Gradient of Knowing model integrate cognitive understanding, controlled tryout, real-world practice, mastery, creativity and field-building into a useful learning-progression framework and assessment rubric?

The answer offered here is provisional. GoK-8 has conceptual value as a framework and rubric candidate. It is not yet a validated theory. The purpose of this paper is therefore to define the model, locate it against established frameworks, identify its limitations and outline a validation agenda.

2. Related Literature

2.1 Cognitive Objectives

Bloom's taxonomy and the revised taxonomy provide a widely used language for educational objectives. Their main value for GoK-8 is in distinguishing early cognitive evidence: recognition, recall, understanding, application and creation. They help describe why factual knowledge is not the same as explanation or creative production.

However, cognitive taxonomies do not by themselves solve the problem of accountable performance. A learner can meet a cognitive objective in a classroom setting while still lacking the ability to act in real conditions. GoK-8 therefore treats cognitive evidence as necessary but not sufficient for practice-oriented competence.

2.2 Competence Assessment

Miller's pyramid is central to the GoK-8 distinction between controlled tryout and accountable practice. The broad Miller distinction between knowing, knowing how, showing how and doing supports the idea that evidence changes when assessment moves from knowledge to demonstration to real performance. In GoK-8, Controlled Tryout / Demonstration describes bounded, guided or simulated performance. Accountable Practice describes performance in real contexts where constraints, stakeholders and consequences are present.

This distinction is one of the most important practical contributions of GoK-8. Many training systems overvalue simulated demonstration because it is easier to observe and score. GoK-8 makes that limitation explicit: demonstration is useful evidence, but it is not the same as accountable practice.

2.3 Skill Acquisition And Mastery

Skill-acquisition models such as Dreyfus and Dreyfus and Fitts-Posner provide comparison points for movement from rule-guided action toward fluent performance. These models support the general idea that competence can change qualitatively through experience and feedback. They also support the distinction between early controlled action and later reliable performance.

At the same time, stage-model critiques require caution. Professional development is not simply a universal movement through fixed stages. It is contextual, embodied, domain-specific and often nonlinear. A person may be a master in one domain and a beginner in another. A learner may show advanced understanding but weak practice, or strong routine performance but limited adaptivity. GoK-8 therefore should be read as an evidence map, not as a deterministic ladder.

2.4 Structural Understanding

SOLO taxonomy is useful for defining Understanding. In GoK-8, understanding is not treated as confidence, fluency or repetition. It requires observable structural evidence: explaining relationships, comparing cases, giving reasons and transferring explanation to nearby examples. This helps avoid a common assessment problem: mistaking fluent vocabulary for conceptual structure.

2.5 Adaptive Expertise And Creativity

The transition from Mastery to Creative / Adaptive Expertise is a major conceptual boundary. Routine mastery means stable, reliable performance across varied cases. Adaptive expertise requires more: the ability to modify methods, transfer across contexts and create novel solutions while preserving quality.

For this reason, GoK-8 does not equate creativity with novelty alone. A new solution can be poor, irrelevant or inefficient. Creative evidence must show adaptive fit, conceptual grounding and quality. This distinction is important in scientific work, software engineering, design, education and professional practice.

2.6 Learning Progressions And Rubrics

Learning-progression literature supports cautious positioning. A proposed progression needs more than an intuitive sequence. It needs conceptual analysis, empirical recovery, comparison to competing models and evidence of instructional usefulness. For GoK-8, this means the current paper can propose a framework, but it cannot claim that learners empirically move through these levels as a universal pattern.

Rubric literature adds another constraint. Moskal and Leydens emphasize that rubric validity depends on the intended purpose of assessment and on whether scoring criteria address relevant content, constructs and criterion-related performance. They also note that a well-designed rubric cannot compensate for a poorly designed assessment task. Reliability requires attention to scorer consistency, clear categories, anchor examples and rater discussion. McHugh's discussion of kappa reinforces the need to measure agreement when multiple raters classify evidence. Percent agreement can be useful but may overestimate agreement because it does not account for chance.

These sources imply that GoK-8 cannot be considered a valid or reliable assessment tool merely because the level labels are plausible. Validation must be designed and executed.

3. The GoK-8 Framework

GoK-8 interprets knowing through four assessment dimensions:

- cognitive structure: what the learner can recognize, explain and relate;
- performance context: whether action occurs in recall, simulation, real-world work or variable contexts;
- accountability: whether outcomes have consequences for stakeholders;
- generativity: whether the learner can adapt, create, teach, standardize or build durable practice.

The framework is summarized in Table 1.

Level	Label	Primary evidence	Typical artifact	Boundary condition
1	Unawareness	Does not recognize the domain, task, vocabulary or relevance.	Blank or inaccurate pre-assessment; inability to name basic terms.	Absence of evidence is not always unawareness; access and language barriers must be checked.
2	Knowledge	Recognizes terms, recalls facts and repeats definitions.	Quiz, glossary, notes, identification task.	Recall is not yet explanation or use.
3	Understanding	Explains relationships, compares cases and reasons about consequences.	Concept map, explanation, worked example with rationale, oral defense.	Confidence or fluent language is not sufficient without structural evidence.
4	Controlled Tryout / Demonstration	Performs a bounded task under guidance, model, simulation or reversible conditions.	Lab task, sandbox exercise, simulation, supervised demonstration, prototype by example.	Demonstration does not prove accountable real-world practice.
5	Accountable Practice	Applies competence in a real context with constraints, stakeholders and consequences.	Real project work, client/user task, production artifact, field performance log.	One successful real case does not prove stable competence.
6	Mastery	Performs reliably across varied contexts, diagnoses problems and requires little supervision.	Portfolio of repeated successful work, expert review, incident handling, mentoring evidence.	Routine fluency can become rigidity if adaptation is weak.
7	Creative / Adaptive Expertise	Modifies methods, transfers across contexts and creates novel solutions while preserving quality.	New method variant, cross-domain solution, original design, validated improvement.	Novelty alone is not creativity; adaptive fit and quality are required.
8	Founding / Field-Building	Creates durable practices, methods, tools, standards or communities adopted by others.	Published framework, adopted standard, community of practice, repeatable method used by others.	Provisional trans-individual extension; not a normal individual learning level.

Levels 1-7 describe individual learning and performance evidence. Level 8 is different. It depends on adoption by others, not only on what the individual can do. A person can be creative without founding a field or community. Founding requires a social trace: methods, standards, tools, schools or practices that persist beyond the individual act.

4. Proposed Validation Agenda

GoK-8 requires validation before it can be used as an assessment instrument. The validation agenda should proceed in stages.

First, expert content review should examine whether the level descriptors are clear, relevant and non-overlapping. Experts should include people with experience in learning assessment, professional development and the selected pilot domain. They should evaluate whether each level has meaningful indicators and whether any level mixes multiple constructs.

Second, a pilot domain should be selected. Scientific-writing development is proposed as the first domain because it produces observable artifacts across levels. A novice may be unable to define basic research terms. A learner may collect sources but fail to synthesize them. Another may draft an introduction under guidance. A more advanced writer may produce a full manuscript with accountable claims, source verification and revisions. A highly developed writer may create new frameworks or methods that others adopt.

Third, an artifact sample should be created. Possible artifact types include notes, literature matrices, research briefs, outlines, manuscript drafts, revision histories, response-to-reviewer tables and submission packages. The sample should remove personal or sensitive information.

Fourth, raters should be trained. At least two independent raters should score calibration artifacts, discuss disagreements and refine the scoring guide before formal scoring. Anchor examples should be created for each level where possible.

Fifth, inter-rater reliability should be estimated. If each artifact receives one ordinal level, weighted kappa may be appropriate. If raters assign numeric scores across dimensions, an intraclass-correlation approach may be considered. Percent agreement can be reported descriptively, but it should not be the only reliability evidence.

Sixth, GoK-8 should be compared to competing models. Possible alternatives include a Bloom-only classification, a Miller-only competence model or a simpler four-level progression. If GoK-8 does not improve interpretation beyond these alternatives, its scientific value is weaker.

Finally, instructional usefulness should be tested. A framework is more valuable if it improves feedback, self-assessment, mentoring decisions or revision quality. This is especially important if GoK-8 is used as a formative tool rather than a grading instrument.

5. Discussion

GoK-8's strongest contribution is integrative. It brings together cognitive objectives, competence assessment, skill acquisition, structural understanding, adaptive expertise and rubric validation into one practical map. Its most useful distinction may be between Controlled Tryout / Demonstration and Accountable Practice. This distinction is common in professional life but often weakly represented in ordinary educational language.

The framework also clarifies the difference between mastery and adaptive creativity. A master can perform reliably in many situations. A creative or adaptive expert can modify methods and create fit-for-purpose solutions when routine patterns are insufficient. This matters in domains where change is constant, such as scientific work, software engineering, organizational development and design.

The model is also intentionally cautious about field-building. Founding / Field-Building is not simply "being very good." It requires adoption evidence by others. This makes the level harder to assign, but it prevents premature self-attribution. A person cannot simply claim to be at level 8; the evidence must include durable uptake.

There are important risks. First, GoK-8 may be mistaken for another rigid stage model. This paper rejects that interpretation. Levels are heuristic descriptors of evidence. They may vary across domains, and learners may show mixed profiles across dimensions. Second, GoK-8 may appear too similar to existing models. Its defense depends on whether the integration improves assessment language and development decisions. Third, without empirical validation, GoK-8 remains a conceptual framework and rubric candidate.

The current paper is therefore best read as an invitation to validation. It defines a model, identifies its relationship to existing literature and makes its limitations explicit.

6. Limitations

This paper does not present empirical validation. It does not show that learners move through GoK-8 levels in a fixed order. It does not establish inter-rater reliability. It does not prove predictive validity or instructional effectiveness.

Several source claims remain access-limited. Some foundational sources are verified through bibliographic records or publisher metadata but still require full-text close-reading before final journal submission. The current draft avoids detailed claims from sources that were not fully accessible in this pass.

The level labels also require further refinement. Controlled Tryout / Demonstration is a workable English label for level 4, but it may need terminological testing. Founding / Field-Building remains the most speculative

descriptor and should be treated as a trans-individual extension rather than a normal individual learning level.

Finally, the proposed pilot domain, scientific-writing development, is convenient and artifact-rich, but it may not generalize to other domains. GoK-8 should be tested in several domains before broad claims are made.

7. Conclusion

GoK-8 offers a structured vocabulary for distinguishing forms of knowing that are often conflated: recognition, explanation, controlled performance, accountable practice, mastery, adaptive creativity and field-building contribution. Its current value is conceptual and practical. It can help identify what kind of evidence is present, what kind is missing and what transition blockers may prevent development.

The next step is validation. A scientific-writing pilot can test whether trained raters can apply the rubric consistently, whether GoK-8 adds interpretive value beyond simpler models and whether GoK-8-guided feedback improves development decisions. Until then, GoK-8 should be presented as a proposed framework and validation agenda, not as a validated theory.

Declarations

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